# Bartłomiej Hawryło

Available for Remote AI Training & Engineering

**Software Engineer | AI & LLM Systems Developer | Full-Stack & XR Creator**

bartlomiejhawrylo@gmail.com · +48 512 037 518 · Gdańsk, Poland (CET · Remote Worldwide) · github.com/barrtk
Portfolio: barrtk.github.io

## PROFESSIONAL SUMMARY

**Versatile Software Engineer & AI Systems Developer** with 10+ years of self-driven programming across **Python, JavaScript/TypeScript, C++, and C#**, combined with a formal **Master of Science in Psychology**. Proven track record architecting complex multi-language AI pipelines: bidirectional engine-to-LLM reasoning bridges (Stockfish + LLMs), real-time multi-directional live translation platforms, local AI voice cloning/TTS dubbing, computer vision pipelines (OpenCV, MediaPipe), and autonomous bots. Possesses a deep, dual-competency in **rigorous algorithmic debugging** and **human cognitive evaluation**—ideal for **RLHF evaluation, LLM reasoning alignment, code quality benchmarking, and AI model training at Outlier AI**.

## TECHNICAL SKILLS & COMPETENCIES

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|---|---|
| **PROGRAMMING LANGUAGES** | `Python (AsyncIO, NumPy, OpenCV)` `TypeScript / JavaScript (ES6+, Node.js)` `C++ (Modern, Unreal Engine)` `C# (.NET Desktop)` `HTML5 & Vanilla CSS` |
| **AI, LLMS & RLHF EVALUATION** | `LLM-to-Engine Logic Bridges` `Chain-of-Thought (CoT) Prompting` `RLHF & Model Hallucination Auditing` `Local Model Inference (Ollama/GGUF)` `Voice Cloning & TTS Dubbing` `Maia Chess Neural Nets` |
| **COMPUTER VISION & XR** | `OpenCV` `MediaPipe Hand Landmark Tracking` `Neural Face Synthesis & Facial Landmark Mapping` `Tesseract OCR & Screen Parsing` `Three.js / WebGL` `WebXR & OpenXR` |
| **FULL-STACK & SYSTEMS** | `Real-Time WebSockets` `REST APIs` `Low-Latency Streaming` `Desktop Automation (Win32/C#)` `Reverse Engineering` `Git & CI/CD` |

## FLAGSHIP AI & SOFTWARE ENGINEERING PROJECTS

### ReThink Chess (ChessTrainer) — AI-Powered Chess Suite (Prototype)

2023 – 2025

Archived Prototype (rethink.fast) · GitHub: @barrtk

Stack: JavaScript (ES6+), Three.js (WebGL), Stockfish Engine (UCI/WebAssembly), Maia Neural Network, LLM Integration, OCR

- **Engine-to-LLM Natural Language Bridge:** Architected a translation layer converting raw Stockfish engine evaluations, multi-ply principal variation lines, and tactical motifs (pins, skewers, forks) into clear, pedagogical human reasoning using LLM prompt pipelines.
- **Proprietary "Rainbow Mode" Visual Layer:** Engineered real-time algorithmic board analytics rendering dynamic color-coded visual overlays to instantly communicate square control, piece vulnerabilities, and tactical tension.
- **Neural Network Integration:** Deployed Maia chess neural models to simulate human-like decision tiers rather than pure cold minimax calculations, enabling granular cognitive training.
- **Computer Vision Screenshot Import:** Built a drag-and-drop OCR pipeline to detect board orientation, piece coordinates, and FEN state directly from image uploads.

### Poliglota Network & Japan-Master — Real-Time Translation & EdTech

2023 – Present

Live: poliglota.network · poliglota.network/japan

Stack: TypeScript, JavaScript, WebSockets, Real-Time Translation APIs, Mobile-First Responsive PWA

- **Multi-Directional Live Translation Matrix:** Designed an N-way real-time communication platform where each participant independently sets their native spoken/typing language and desired reading language, routing messages with sub-second latency.
- **Japan-Master Mnemonic Engine:** Built an active-recall web app applying visual mnemonics to master 1,000+ Japanese Kanji characters (**poliglota.network/japan**).

### Stremio-Lektor — Local AI Voice Cloning & Dual Subtitles

Custom Stremio Engine Addon

2024

Stack: JavaScript, Node.js, Local TTS & Voice Cloning AI, SRT/VTT Stream Parser, Japanese Romaji Phonetics

- **Automated AI Voice Dubbing:** Integrated local neural voice-cloning models to synthesize synchronized audio dubbing directly from streaming subtitle tracks on-the-fly.
- **Dual Phonetic Subtitling:** Developed an automated parser translating subtitle streams into dual-language Japanese with inline Romaji phonetic transcriptions for language immersion.

## SYSTEMS, COMPUTER VISION, XR & AUTOMATION PROJECTS

**SwissKnife Toolbox (SKToolbox) — Desktop OCR & Automation** Live at sktoolbox.pro 2024 – Present

Stack: C#, .NET WinForms/WPF, Windows API Hooks, Tesseract OCR Engine, Clipboard Streaming

- Developed and shipped a production Windows desktop utility providing near-instant region screen capture, automated OCR text extraction, and direct clipboard/chat pipe integration.
- Optimized for low-overhead memory consumption and hotkey responsiveness, used daily across real-world workflows.

**RemoteFaceApp & Gesture Vision — Real-Time CV Pipelines** Computer Vision Research & Development 2023 – 2024

Stack: Python, OpenCV, MediaPipe, Neural Face Synthesis Pipeline, Socket Low-Latency Streaming

- **RemoteFaceApp:** Built a low-latency PC-to-mobile remote desktop control tool featuring real-time neural face synthesis and facial reenactment integrated into the video transmission stream.
- **Gesture Tracking:** Implemented real-time multi-point hand landmark detection and gesture spatial mapping using MediaPipe and OpenCV for touchless software interaction.

**Autonomous Game Agent — Documented World Record (Floor 13,479)** 2026

Verified 1h 60FPS Demonstration on YouTube

Stack: Python, Memory/Pixel Frame Analysis, Real-Time Microsecond Input Automation, State Machine

- Programmed an autonomous bot for Icy Tower that successfully reached Floor 13,479, surpassing all documented internet records through millisecond-precision momentum calculation and dynamic ledge detection.

**VR & WebXR Piano Learning Suite (VrPiano554 & openxr_piano)** C++ / Unreal Engine 5.5.4 & Three.js WebXR 2024

Stack: C++, Unreal Engine 5.5, OpenXR, Three.js, WebGL, MIDI Hardware Protocols

- Engineered an interactive spatial piano learning tool (similar to PianoVision) with falling 3D note trajectories mapped directly onto physical MIDI keyboards in real-time, achieving zero perceived latency.

## WORK EXPERIENCE

**Independent AI & Full-Stack Developer** 2022 – Present

Self-Employed / Web & Desktop Projects

- Designed, deployed, and maintained production web apps (rethink.fast, poliglota.network, sktoolbox.pro).
- Engineered custom LLM prompt architectures, voice synthesis models, and computer vision systems.

**Transport Psychologist & Case Analyst** 2026

dobrebiuro.de (Fully Remote)

- Conducted rigorous diagnostic interviews and behavioral case evaluations for drivers undergoing German MPU assessments.
- Applied systematic analytical frameworks to detect inconsistencies and evaluate cognitive reliability.

**Customer Operations Specialist** 2025

Telus International · Gdańsk

- Handled multi-channel support within enterprise CRM & database environments under strict SLA standards.

## EDUCATION & CREDENTIALS

**M.Sc. in Psychology** 2019 – 2024

WSB Merito University, Gdańsk

Focus: Cognitive processes, diagnostic methodology, behavioral analysis, and human decision-making.

**Postgrad: Transport Psychology** 2025 – 2026

SWPS University, Sopot

Official license as transport psychologist (Reg. No. G-P 359).

**IT Technician Diploma (CS)** 2012 – 2016

Technical High School No. 1, Sopot

Curriculum: Algorithms, operating systems, networking, database administration, and OOP.

### LANGUAGES

**English:** Full Professional Working Proficiency (C1)

**Polish:** Native

**Why Bartłomiej is Uniquely Qualified for Outlier AI**

Combines **cross-language coding execution (Python, JS/TS, C++, C#)** with an **M.Sc. in Psychology**. Uniquely equipped to evaluate both technical code correctness (algorithmic edge cases, execution performance, security bugs) and human-cognitive alignment (Chain-of-Thought validity, hallucination detection, instruction adherence).

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